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Toward realistic and lightweight intrusion detection for ADS-B: A comprehensive threat model and multi-attack machine learning evaluation

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ABSTRACT

Automatic Dependent Surveillance–Broadcast (ADS-B) is a critical surveillance technology for modern air traffic management. However, the protocol's open and unencrypted nature exposes it to significant cybersecurity threats. This paper presents a comprehensive and realistic intrusion detection system (IDS) for ADS-B based on machine learning (ML) models. Existing research has primarily focused on limited attack scenarios or binary classification (benign vs. malicious). In contrast, this study introduces a multi-attack evaluation framework covering nine distinct ADS-B attack types under realistic traffic conditions. A novel dataset is constructed by combining real-world ADS-B data from the OpenSky Network with simulated attack scenarios generated using an enhanced OpenScope platform. The dataset includes diverse attack types such as ghost injection, trajectory manipulation, spoofing, and message delay. Furthermore, a structured threat model based on STRIDE and CIA+A principles is proposed to systematically map vulnerabilities to attack behaviors. Three ML models: KNN, SVM, and XGBoost—are implemented and evaluated using multiple performance metrics, including accuracy, precision, recall, F1-score, MCC, ROC-AUC, and PR-AUC. Experimental results show that both KNN and XGBoost achieve high classification accuracy (98.16%) and (98%), significantly outperforming SVM (83%). KNN demonstrates strong macro-level performance (F1-score: 0.95) and low computational overhead, making it suitable for near real-time deployment. The real-time feasibility analysis indicates that the proposed system can support high-density air traffic scenarios through parallel processing. The findings highlight the effectiveness of lightweight ML-based IDS approaches for securing ADS-B communications, while also identifying challenges in detecting temporally subtle attacks such as message delay. To the best of our knowledge, this work is among the first to jointly evaluate multi-attack detection, dataset realism, and real-time deployment feasibility within a unified ADS-B intrusion detection framework.

1. Introduction

During recent years, air travel has experienced significant growth due to ongoing economic development. As per the report of the Federal Aviation Administration (FAA), the number of passengers is expected to reach 1.15 billion by 2030 [1]. Airlines have increased flight operations to meet the growing demand, resulting in higher air traffic density and more congested airspace. To solve the growing flight operations and complexity of modern air traffic management (ATM), in 2005, the Department of the FAA of the United States (US) introduced the Automatic Dependent Surveillance Broadcast (ADS-B) protocol. To enhance airspace safety and efficiency, the protocol was integrated as a core component of the Next Generation Air Transportation System (NextGen) project.

The FAA makes ADS-B mandatory for aircraft across the US and Europe and is increasingly being deployed in large commercial drones and military Unmanned Aerial Vehicles (UAVs). The protocol enhanced passenger safety and collision avoidance, increased airspace situational awareness, and facilitated the safe integration of drones and UAVs into shared airspace with manned aircraft. Additionally, the continuous growth of drones in commercial and agricultural domains, as well as in surveillance and military operations, has also introduced complexity to ATM systems.

Existing studies show a significant increase in low-altitude airspace density and coordination complexity due to the deployment of UAVs in agriculture, logistics, and infrastructure monitoring [2]. The integration of unmanned traffic management (UTM) with traditional ATM systems has therefore faced critical safety challenges, specifically

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for beyond-visual-line-of-sight (BVLOS) operations [3,4]. Considering the identified future challenges, an innovative and advanced security solution is required to ensure the airspace is safe and efficient [5].

ADS-B and Controller Pilot Data Link Communication (CPDLC) protocols are widely deployed for effective and efficient air traffic communication. Since 2020, ADS-B has been mandatory for commercial aircraft and played a vital role in modern air traffic surveillance, and it is a key part of the FAA's NextGen project. The purpose of the project was to improve crowded airspace surveillance [6]. The protocol belongs to the class of Electronic Conspicuity (EC) technologies. The class also includes FLARM and MLAT technologies for airspace surveillance, sense-and-avoid capabilities, and mid-air collision prevention. EC has now been expanded to include unmanned and mixed-traffic airspace. Security vulnerabilities in ADS-B directly affect the reliability of EC-based situational awareness, underscoring the need for security mechanisms that operate without protocol modification.

The advantages of the protocol include accurate GPS-based location tracking, cost-effectiveness, easy deployment, and improved situational awareness for pilots and ATC. The protocol also has several weaknesses in addition to its advantages. The protocol lacks built-in security features, including encryption and authentication [7]. The lack of built-in security features makes the protocol vulnerable to several cyberattacks, including spoofing, jamming, eavesdropping, and message tampering [8].

Existing studies show that with tools such as Software-Defined Radios (SDRs), even individuals with sufficient technical skills can exploit these vulnerabilities [9]. Two fundamental mitigation solutions have been explored in existing literature to address the identified cyberattacks: cryptographic-based solutions and AI-based anomaly detection systems.

Cryptographic-based solutions, including encryption, digital signatures, and authentication codes, can secure ADS-B, but these solutions require modifications in the ADS-B message structure, which can be tough to implement across a global network [10–12]. On the other hand, AI-based solutions focus on suspicious patterns in air traffic data without requiring changes to the protocol structure. These solutions can identify unusual flight behaviors, including unexpected transmissions, thereby helping to detect threats in real-time [13,14].

This paper differentiates itself from existing research, including our own prior work [15] in the following key aspects.

- First, the paper comprehensively explains and visualizes the generation process of a multi-attack dataset containing nine ADS-B attack types, significantly expanding beyond existing work that typically considers limited or single-attack scenarios.
- Second, a structured threat model for the ADS-B protocol is developed by combining the STRIDE model and CIA+A security principles, enabling systematic mapping of security requirements, threats, and attack behaviors.
- Third, the paper highlights a lightweight, real-time IDS using three ML models including SVM, XGBoost, and KNN that achieves high detection accuracy without relying on computationally expensive deep learning models.

These contributions collectively address key gaps in realism, scalability, and operational applicability identified in prior ADS-B security research.

1.1. Paper organization

The rest of the paper is organized as follows: Section 2 discusses the existing ML and DL-based security solutions and the strengths and weaknesses of the studies. Section 3 presents a novel ADS-B threat model and maps the identified attacks based on the threats and CIA+A, whereas Section 4 discusses the methods used for dataset generation and the limitations of existing datasets. Section 5 outlines the proposed research methodology, including the flowchart of the developed IDS, dataset generation, and preprocessing steps. Section 6 presents the experiments and results of the implemented models along with their evaluation metrics, whereas Section 7 discusses the performance of the implemented models. Section 8 concludes the paper by highlighting future research directions and the study's limitations.

2. Literature review

ADS-B is an important component of the current era of the aviation industry, providing essential real-time tracking and communication capabilities. Although ADS-B is widely adopted, its inherent vulnerabilities pose significant cybersecurity challenges. In recent years, several security solutions have been developed to mitigate these risks, addressing different facets of the ADS-B ecosystem. These solutions include:

- **Anomaly Detection with AI:** AI, including ML and DL methods, has emerged as an effective instrument for identifying harmful ADS-B messages. Techniques such as Convolutional Neural Networks (CNNs), Long Short-Term Memory (LSTM) networks, and ensemble learning models leverage historical and recent data to identify various attacks with high precision.
- **Cryptographic Methods:** Strategies designed to maintain data integrity and confidentiality, yet these methods are constrained by resource limitations and legacy systems.
- **Authentication via Blockchain:** Distributed methods aimed at improving the trustworthiness of ADS-B communications.
- **Fingerprinting Techniques:** Methods for recognizing transmitters by employing distinct signal traits to differentiate between benign messages and malicious ones.
- **Location/Time Validation Methods:** Techniques for validating ADS-B data through checks of temporal or spatial consistency.
- **Kalman Filter-Driven Approaches:** Methods for estimating trajectories and identifying anomalies in flight trajectories.
- **Signal and Physical Layer Improvements:** Strategies to enhance communication resilience against jamming and interference.

This paper focuses exclusively on ML- and DL-based approaches, as existing solutions have already been systematically categorized and discussed in our recent survey [16]. Here, we identify the strengths, limitations, and applicability of ML and DL-based solutions for securing ADS-B communication.

In [17], the authors demonstrated the efficacy of ML-based methods in classifying ADS-B messages as benign and malicious and identifying specific attack types. The research has shown that Random Forest (RF) models achieved the highest accuracy for binary classification and multi-class classification. The experiments were performed on the dataset containing 22,316 messages (50% malicious and 50% benign). On the other side, in [18], the authors used ensemble meta-learning methods using stacked random forest and XGBoost with logistic regression meta-learner combined with Explainable AI (XAI). The authors focused on detecting Ghost Aircraft Injection, Path Modification and Velocity Drift attacks. For the experiments, the authors utilized the dataset published by [19]. In [20], the authors reviewed supervised and unsupervised ML techniques to detect anomalies in ADS-B communication. The study focused on designing robust models capable of classifying and predicting anomalies effectively. An anomaly detection system for ADS-B data was designed by [21], targeting enhanced detection of various attack scenarios using a hybrid dataset. The study evaluated the effectiveness of ML and DL models in detecting anomalies in ADS-B data.

In [22], the authors introduced an ML-based approach tailored to realistic low-altitude ADS-B attack detection by implementing four specific attack scenarios: spoofing, saturation, replay, and interpolated ghost trajectories. The authors developed detection approaches that leverage ADS-B time-series context to enhance algorithm performance, using a Python-based anomaly-detection library applicable to any ADS-B data source. Similarly, in [23], the authors developed a real-time-capable, computationally efficient ML-based IDS for ADS-B to detect and classify injection attacks. The ADS-B Message Injection Attacks Dataset is used for training and evaluation.

In [24], the authors presented an ML-based IDS for ADS-B communication that targets jumping aircraft, false information display, virtual trajectory modification, and transponder code alteration attacks. The authors in [25] developed an ML-based module within the ADS-Bsec framework to classify ADS-B attacks. This study emphasized the multi-class classification of attack types (replay attacks, ghost aircraft injection, and multiple ghost aircraft injection). The best performance was achieved by the Decision Tree model. A limitation of these studies is the narrow diversity of attack types.

In [26], the authors evaluated various DL architectures, including Long Short-Term Memory (LSTM) and its variants, for detecting anomalous ADS-B messages. It focused on detecting False Data Injection Attacks (FDIAs) that alter altitude values in ADS-B messages. Models like LSTM and Gated Recurrent Units (GRU) excel at detecting Path Modification, Ghost Aircraft Injection, and Velocity Drift by capturing temporal dependencies in ADS-B data [27]. Transformer-based models have also shown exceptional performance in sequential anomaly detection for replay attack detection using a dataset that consisted of ADS-B data from a flight between Lisbon and Paris [28]. In [29], the authors proposed the TTSAD (TCN-Transformer-SVDD) model for anomaly detection in ADS-B data by leveraging temporal correlation and distribution characteristics, ensuring efficient and accurate detection of attacks such as random position deviation, fixed position deviation, and Denial-of-Service (DoS) attacks. The performance validation of the proposed model is limited to simulated attack scenarios without testing on diverse real-world datasets.

In [30], the authors aim to enhance ADS-B data classification, specifically for identifying aircraft engine types, by optimizing dataset size and feature selection to improve predictive performance under limited computational resources. Authors combined LSTM, FCN, and squeeze-and-excite blocks for improved multivariate time-series classification. An LSTM-based method for detecting spoofing attacks on ADS-B messages was proposed by [31]. It aims to identify manipulated features in ADS-B data by applying sliding windows for temporal preprocessing and residual-based anomaly thresholds. The LSTM model detected various spoofing attacks.

In [32], the authors proposed a novel GNSS interference detection method for ADS-B systems using DL models, such as RNN and LSTM, and a multi-index feature evaluation system. The aim was to enhance detection accuracy by incorporating spatiotemporal information through sliding window data construction. The LSTM model achieved the highest accuracy of 96.13%. In [33], the authors reviewed ML and DL techniques for anomaly detection in ADS-B systems. The authors identify effective models, evaluate their advantages and disadvantages, and highlight challenges and opportunities for securing ADS-B communication. The authors identify that LSTM often achieves high accuracy in detecting trajectory deviations, and SVM effectively detects jamming and spoofing attacks.

A significant drawback of these approaches is the complexity of DL models, which can result in increased computational overhead and longer training times. They are also highly sensitive to noise in the data. The literature shows that ML-based methods are not only less computationally demanding but also achieve better results than their DL-based counterparts on ADS-B data. Table 1 summarizes the most recent ML and DL-based solutions. After a thorough critical analysis of the state-of-the-art literature, we identified the following research gaps:

- Development of a new threat model: There is a need for a comprehensive threat model tailored to ADS-B. This model should systematically identify ADS-B attack scenarios and associate them with specific security requirements.
- Comprehensive dataset for ADS-B security: The lack of a publicly available comprehensive dataset containing diverse and realistic ADS-B attack scenarios hampers the development and evaluation of IDS. A well-structured dataset encompassing various attack types is essential to strengthen ADS-B protocol security.
- Lightweight IDS for ADS-B: An effective and lightweight IDS is required for the classification of ADS-B messages and for accurately identifying specific attack types to mitigate security risks.

3. ADS-B threat model

A threat model, also referred to as an adversary model, provides a structured methodology for identifying, analyzing, and mitigating security risks and vulnerabilities within a system, protocol, or application. To systematically characterize the security threats affecting ADS-B, this work adopts the Microsoft STRIDE threat modeling framework [37] as the foundation for the proposed ADS-B threat model. STRIDE categorizes threats into six classes: Spoofing, Tampering, Repudiation, Information Disclosure, Denial of Service, and Elevation of Privilege, each representing a distinct attacker objective.

ADS-B was designed as a cooperative, broadcast-based surveillance protocol and does not incorporate built-in security mechanisms such as encryption, authentication, or integrity protection. As a result, confidentiality and message repudiation are not treated as primary security objectives in this threat model. Instead, the model focuses on threats that exploit the lack of authentication, integrity verification, and availability guarantees. In particular, it is assumed that an adversary may exploit ADS-B vulnerabilities to:

- Passively intercept broadcast communications, enabling information disclosure;
- Inject fabricated messages or alter transmitted data, compromising message integrity and authenticity;
- Disrupt the communication channel, degrading system availability through interference or jamming.

In operational environments, ADS-B position estimates are affected by GNSS error sources including dilution of precision (DoP), ionospheric delay, receiver clock bias, and satellite geometry. These effects introduce natural noise into latitude, longitude, and altitude fields, which may partially mask or mimic deviations. While Kalman filter-based approaches have been proposed to smooth such noise and detect trajectory anomalies, they often assume Gaussian error models and require continuous temporal context, limiting their applicability for lightweight, per-message IDS deployment.

As illustrated in Fig. 1, these threats affect both the air and ground segments of the ADS-B ecosystem and include spoofing, message injection, message modification, eavesdropping, and jamming. Ground-based attackers operate from terrestrial locations using equipment such as Software-Defined Radios (SDRs) to transmit, inject, or manipulate ADS-B messages received by ground infrastructure. If falsified or modified data is ingested by ATC systems, it may distort the ATC's situational awareness, potentially leading to misrouting decisions, the appearance of ghost aircraft, or traffic separation conflicts. Air-based attackers operate from airborne platforms, such as malicious or compromised aircraft, and can transmit counterfeit ADS-B messages while impersonating legitimate aircraft or altering position, velocity, or identification fields. Due to their proximity to other aircraft, such attacks may have an immediate impact on airborne situational awareness and collision avoidance systems.

The proposed threat model captures the core entities and data flows that define the ADS-B operational environment. Aircraft determine their position, speed, and heading using Global Navigation Satellite System (GNSS/GPS) data. This information is periodically broadcast via ADS-B Out to nearby aircraft and ground-based receivers to support cooperative surveillance. Because these broadcasts are unencrypted,

Table 1
Summary of most recent ML and DL-based solutions for protecting ADS-B from cyber attacks.

Ref.	Research objectives	Dataset details	Models with accuracy	Limitation
[25]	Classify ADS-B attack types, including ghost aircraft injection and replay attacks.	Simulated ADS-B data with distributions of benign messages (95–05, 90–10, 85–15).	RF: 99%, DT: 98%, SVM: 97%.	It is limited to simulated attack scenarios and lacks scalability testing for real-time applications.
[24]	Detect intrusion types such as jumping aircraft, false information, and trajectory modifications.	OpenSky Network ADS-B data: 26,000 malicious and 25,000 benign messages were simulated for analysis.	KNN: 99.8%, LR: 98%, NB: 96%.	Limited to four attack types; scalability to large-scale air traffic systems not explored.
[31]	Detect spoofing attacks on ADS-B data using LSTM and temporal data preprocessing.	220,000 ADS-B messages from GitHub; simulated spoofing attacks with systematic altitude and velocity changes.	LSTM (F1-scores: 0.30–0.98 depending on attack type).	Accuracy varies across attack types; it lacks real-world validation and testing on diverse datasets.
[34]	Evaluate vulnerabilities of ADS-B anomaly detection models to adversarial attacks.	OpenSky Network ADS-B data with 50 training flights and 40 testing flights, augmented with synthetic anomalies.	LSTM: 98%, GRU: 97%, LSTM-Encoder-Decoder: 95%.	Focused on simulated data, it lacks real-world validation.
[26]	Evaluate DL models for detecting anomalies in ADS-B messages, focusing on altitude manipulation attacks.	OpenSky Network ADS-B messages: Training set with 66,172 messages, test set with 19,827 messages; simulated anomalies generated by altering altitude.	Stacked LSTM: F1-Score = 95.84%, Precision = 93.04%, Recall = 98.82%.	Focused on a single attack type (altitude anomalies); scalability and computational costs of LSTM architectures not evaluated for real-time environments.
[30]	Classify aircraft engine types using ADS-B data while reducing dataset size and computational complexity.	ADS-B data from 2020, the training set of 1.23 million samples; features include altitude, speed, vertical speed, latitude, and longitude.	MLSTM-FCN: 89.4%.	Focus on engine classification only; limited geographical and operational coverage.
[21]	Develop an anomaly detection system for various ADS-B attack scenarios.	Hybrid datasets combining eight attack scenarios (e.g., spoofing, virtual trajectory changes).	Random Forest and Extra Tree Classifier: 99.9%, DT: 99.2%.	Focused on hybrid datasets; lacks scalability insights.
[35]	Introduce FEPAN to detect spoofing attacks using frequency-enhanced attention mechanisms.	Simulated ADS-B datasets with deceptive maneuvers across flight phases.	Transformer-based FEPAN: Superior accuracy for spoofing detection (metrics not provided).	Simulation only; lacks validation with real-world datasets.
[17]	Propose a detection method to identify injected ADS-B messages, including zero-day attacks.	ADS-B dataset containing genuine and injected messages.	Random Forest: Binary classification-99.14%, Multiclass-99.41% accuracy.	Limited to specific attack types, a larger dataset is required for better generalization.
[33]	Survey anomaly detection methods in ADS-B systems using ML/DL approaches.	Reviewed datasets in 290 studies; lacks original dataset usage.	ML and DL approaches focus on anomaly detection.	Does not evaluate models or propose new implementations.
[36]	Use GAN-LSTM to detect ADS-B attacks by combining flight plans and ADS-B data to create flight image streams.	Real flight data with synthesized abnormal data simulating various attacks.	GAN-LSTM: Accuracy = 92.3%, False Positive Rate = 11.9%, False Negative Rate = 6.2%.	Limited accuracy for stealthy attacks; focuses on specific ATC-related use cases.
[28]	Detect replay attacks in ADS-B using a Transformer-based model.	Sample ADS-B data from commercial flights.	Transformers: Outperformed other DL models (metrics not provided).	Small dataset; limited to replay attacks only.
[27]	Investigate RNNs for detecting injection attacks in ADS-B data.	ADS-B data with injected false data; details not fully disclosed.	RNN, LSTM, GRU: Evaluated for time-series injection detection (metrics not provided).	Lack of real-time validation and comprehensive dataset description.

unauthenticated, and transmitted over a shared wireless medium, the ADS-B communication channel presents a broad attack surface for both passive and active adversaries.

While several ADS-B threat models have been proposed in the literature, many focus on a limited subset of attacks or adversary capabilities. For example, some of the existing studies concentrate primarily on spoofing attacks [38], while others emphasize denial-of-service [39] or man-in-the-middle scenarios [40]. Although such models provide valuable insights into specific attack vectors, they often do not capture the full range of realistic threats enabled by the broadcast nature of ADS-B. In contrast, the threat model presented in this work offers a more comprehensive and systematic characterization of ADS-B threats, which directly informs the design of the proposed multi-attack dataset and the evaluation of machine learning–based intrusion detection systems.

Fig. 2 illustrates a structured mapping between ADS-B security requirements, threat categories, and concrete attack manifestations. The green boxes represent specific instances of the five high-level threat classes (purple boxes): eavesdropping, spoofing, message modification, message injection, and jamming, each showing how an adversary can operationalize these threats in practice. Examples of such attack manifestations include ghost injection, stationary aircraft, passive eavesdropping, non-responsive aircraft, message delay, and others. These attack instances are grouped according to the security property they violate, namely confidentiality, integrity, availability, and authentication (CIA+A). Although CIA+A has been widely adopted in cybersecurity,

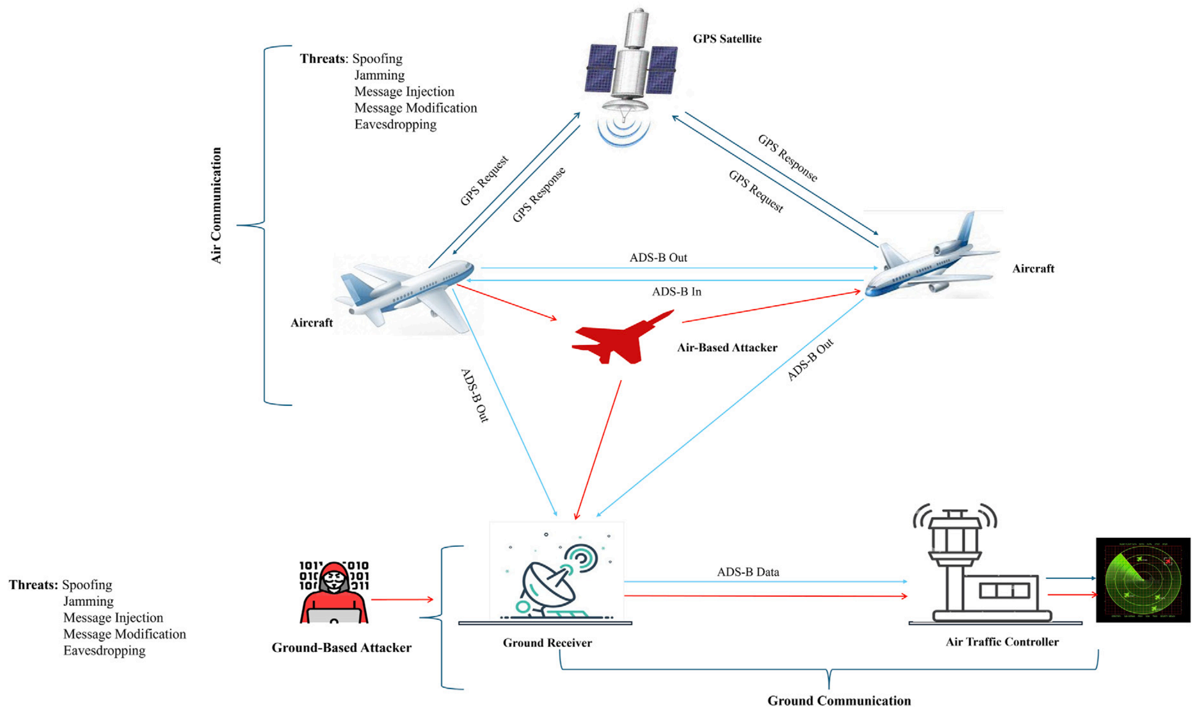

Fig. 1. ADS-B threat model.

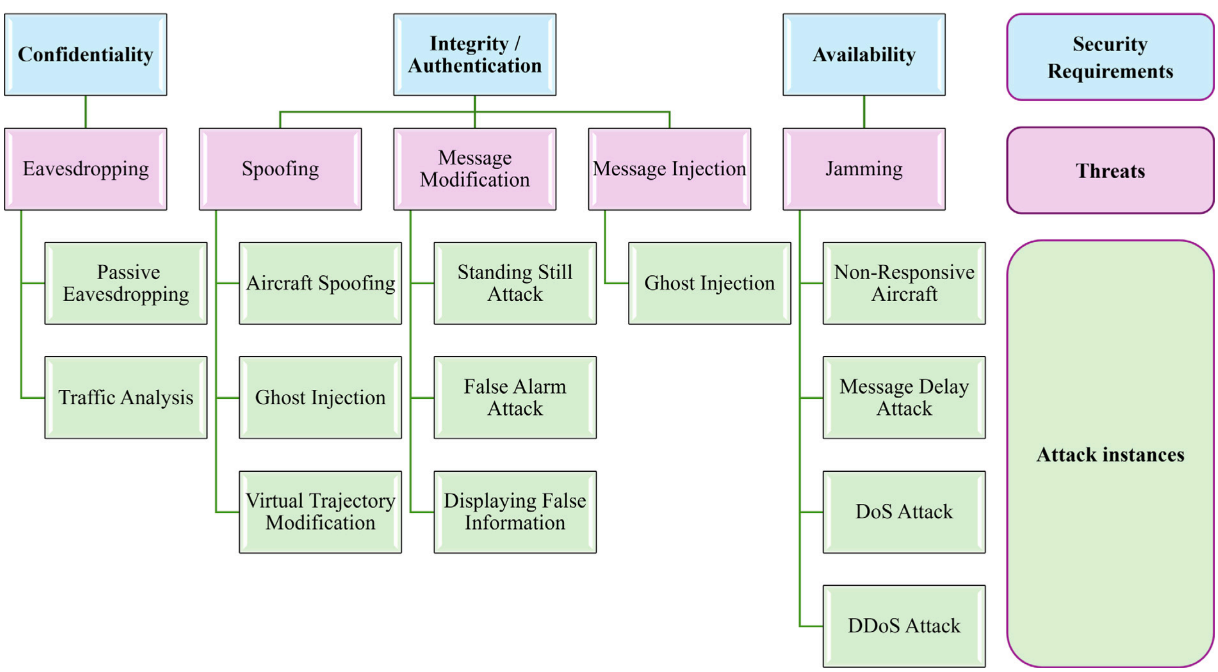

Fig. 2. ADS-B attack classification based on identified security threats and CIA+A.

its application in ADS-B security has largely remained abstract or limited to individual attack types. The novelty of our taxonomy lies in its operationalization for ADS-B, where each security requirement is mapped to specific, observable attack behaviors, which serve as labels in an ML-ready dataset.

Attacks on confidentiality

Attacks on confidentiality exploit the broadcast and unencrypted nature of ADS-B communications, allowing adversaries to passively observe legitimate transmissions. ADS-B messages contain sensitive operational information such as aircraft position, altitude, velocity, identification, and trajectory, which can be collected by any receiver within range [41]. Although ADS-B does not aim to provide confidentiality by design, the unrestricted availability of this information enables subsequent, more advanced attacks.

- Passive Eavesdropping: In passive eavesdropping, the attacker acts solely as a receiver and listens to ADS-B broadcasts without altering them. Because ADS-B messages are transmitted in plaintext and lack authentication, aircraft identifiers (e.g., ICAO addresses), positions, velocities, and routes can be trivially captured.

Information gathered through eavesdropping can later be leveraged to facilitate active attacks, such as aircraft impersonation using previously observed ICAO codes.
- Traffic Analysis: Traffic analysis extends beyond simple message collection by examining communication patterns over time, such as aircraft density in a given airspace, message update rates, inter-message intervals, and recurring flight routes. By analyzing these patterns, an attacker can infer operational behaviors and determine when and where spoofed or injected messages are most likely to appear legitimate. Additionally, knowledge of update intervals enables attackers to synchronize counterfeit broadcasts with real traffic to reduce detectability.

Attacks on integrity and authentication

Integrity and authentication attacks exploit the absence of cryptographic protection and sender verification in ADS-B. While ADS-B messages cannot be reliably intercepted and removed from the airspace, adversaries can inject or rebroadcast falsified messages that overwrite or contradict legitimate data from the receiver's perspective. Since ADS-B messages provide real-time situational awareness to ATC and nearby aircraft, manipulation of this information can result in hazardous operational decisions [42,43].

- Aircraft Spoofing: In aircraft spoofing attacks, adversaries transmit forged ADS-B messages containing falsified flight information such as callsign, position, altitude, or velocity. Due to the lack of authentication in ADS-B receivers, these messages may be accepted as legitimate, misleading ATC systems and nearby aircraft, and potentially causing navigation errors or loss of separation.
- Ghost Injection: Ghost injection involves the creation of non-existent (phantom) aircraft by broadcasting fabricated ADS-B messages. These ghost aircraft appear as legitimate tracks to ATC and pilots, increasing airspace congestion and potentially overwhelming human operators or automated systems.
- Virtual Trajectory Modification: In this attack, adversaries rebroadcast modified ADS-B messages that alter an aircraft's perceived trajectory, such as introducing abrupt turns, altitude changes, or speed variations. Although the attacker does not alter the original transmission, the injected messages create a false motion profile that misrepresents the aircraft's actual behavior.
- Standing Still Attack: This attack forges ADS-B messages indicating that an aircraft is stationary, either on the ground or in mid-air, by manipulating position and velocity fields. While the aircraft is in fact moving, receivers perceive it as static, which can disrupt collision avoidance and traffic flow management.
- False Alarm Attack: In false alarm attacks, adversaries inject ADS-B messages indicating emergency conditions or abnormal aircraft states, triggering unnecessary alerts for pilots and air traffic controllers. Such attacks can increase operator workload and reduce trust in surveillance data.
- Displaying False Information: This attack involves injecting ADS-B messages containing fabricated aircraft identities, positions, or motion vectors, causing receivers to display incorrect or misleading information about airspace conditions.

Attacks on availability

Availability attacks aim to disrupt the timely reception of ADS-B messages, thereby degrading situational awareness. Because ADS-B relies on a shared wireless channel, attackers can disrupt message delivery by jamming or generating benign messages, preventing ATC and aircraft from reliably receiving surveillance data.

- Non-Responsive Aircraft: In this scenario, an aircraft appears non-responsive to ADS-B receivers, either due to deliberate transponder deactivation, signal interference, or disruption. The resulting loss of visibility creates blind spots in air traffic surveillance and increases the risk of mid-air conflicts.
- Message Delay Attack: In a message delay attack, an adversary captures legitimate ADS-B broadcasts and retransmits them after a controlled delay. As a result, receivers process outdated or out-of-sequence position reports, leading to the appearance of duplicate tracks or inconsistent aircraft motion and degrading situational awareness.
- DoS and DDoS Attacks: DoS attacks target ADS-B availability by overwhelming receivers with excessive or malicious traffic or by jamming the ADS-B radio frequency. In Distributed DoS (DDoS) attacks, multiple coordinated transmitters amplify the disruption, making detection and mitigation more challenging. Such attacks can significantly impair the reliability of air traffic surveillance.

4. ADS-B datasets

For the development and evaluation of ML and DL-based IDS, high-quality, representative datasets are essential. In the context of ADS-B security, datasets must contain sufficient volumes of both benign and malicious messages to enable models to learn meaningful behavioral distinctions. ADS-B data is inherently time- and context-dependent; therefore, suitable datasets should capture features such as aircraft position, velocity, message transmission rate, and identifiers over extended time periods. Moreover, to ensure realistic model evaluation, datasets should reflect real-world flight dynamics, communication noise, and irregularities caused by environmental conditions or technical limitations.

4.1. Dataset requirements

To support the training and evaluation of an ADS-B intrusion detection framework, datasets should satisfy the following requirements:

- **Authenticity:** Data should originate from real ADS-B broadcasts or from validated simulation environments that accurately reproduce real-world air traffic behavior.
- **Volume:** A sufficiently large number of message samples is required to support robust ML/DL model training and generalization.
- **Accessibility and Reproducibility:** Datasets should be openly accessible or publicly documented to enable reproducibility and fair comparison across studies.
- **Diversity:** The dataset should encompass a wide range of attack types to enable comprehensive IDS evaluation.
- **Labeling:** Accurate and well-defined ground truth labels (benign and malicious classes) are critical for supervised learning and performance assessment.

4.2. Existing ADS-B data sources and limitations

ADS-B data can be obtained from several sources, including FlightAware, FlightRadar24, ADS-B Exchange, the OpenSky Network, the FAA database, and the OpenScope platform. Table 2 compares these data sources. While most platforms provide authentic real-world ADS-B messages, they typically contain only benign traffic and often require paid subscriptions.

The FAA database offers reliable data but is restricted to U.S. airspace. OpenSky [49] provides both historical and real-time ADS-B data for research and educational purposes, making it one of the most widely used open-access sources. In contrast, FlightAware, FlightRadar24, and ADS-B Exchange generally require paid access for large-scale data retrieval.

OpenScope is currently the only openly available simulation platform that supports the generation of both benign and malicious ADS-B messages. Several studies [50–53] have extended OpenScope by implementing additional attack scenarios, making it particularly valuable

Table 2
Comparison of platforms used for ADS-B datasets generation.

Platforms feature	Type	Data types	Coverage	Paid plans	Free access	API access
FlightAware [44]	Commercial	Real-time and historical	Global	API	Basic features	Yes
FlightRadar24 [45]	Commercial	Real-time and historical	Global	API	Basic features with delay	Yes
OpenSky Network [46]	Open access	Raw	Global	Non-commercial	Research and education	Yes
ADS-B Exchange [47]	Open-source	Raw	Global	Commercial use	Full raw data for personal use	Yes
FAA Database [48]	Government	Historical	U.S.	N/A	Public access	No
OpenScope Network [40]	Open access	Simulated	Global	No	Yes	Yes

for security research. However, simulated environments alone may not fully capture the complexity and variability of real-world flight behavior.

Based on a comprehensive review of the existing literature and available datasets, the following key limitations are identified:

- **Limited availability and reproducibility:** Many studies rely on proprietary or unpublished datasets, limiting reproducibility and hindering longitudinal research.
- **Restricted attack diversity:** Most datasets either contain only benign ADS-B traffic or include a small number of attack types (typically one to four), which constrains multi-attack IDS evaluation.
- **Labeling challenges:** Publicly available datasets with clearly annotated malicious events are scarce, forcing researchers to rely on synthetic labeling or partial simulations.

These limitations highlight the need for more comprehensive, reusable ADS-B security datasets that combine real flight data with well-defined and diverse attack scenarios. To address this gap, this work leverages real-world ADS-B data from the OpenSky Network and augments it with simulated attack scenarios generated using the OpenScope platform, resulting in a diverse, labeled dataset suitable for realistic IDS training and evaluation.

5. Research methodology

This section describes the methodology adopted for detecting and classifying attacks in ADS-B communications. Fig. 3 illustrates the overall research methodology followed in the development of the proposed intrusion detection system (IDS). The methodology consists of five main stages: dataset generation, data preprocessing, model training, model evaluation, and attack classification. In this paper, three models were implemented, including KNN, SVM, and XGBoost. These models were selected for their complementary strengths and effectiveness in classification tasks for ADS-B message analysis and attack detection.

Fig. 4 presents a detailed flowchart of the proposed IDS architecture, which integrates the two core ADS-B functionalities: In and Out. During normal operation, an aircraft determines its precise position using GNSS data and broadcasts this information via ADS-B Out after incorporating additional flight parameters such as velocity, altitude, and heading. These broadcast messages are received by nearby aircraft and ground stations through ADS-B In. The broadcast messages can be analyzed either on board in real time or at ground stations. The IDS continuously monitors incoming ADS-B messages to identify deviations from expected flight behavior. When anomalous activity indicative of an attack is detected, the system generates an alert to trigger further verification or response actions. If no anomaly is detected, the IDS continues monitoring the message stream to ensure continuous surveillance and operational integrity.

5.1. Data collection

The dataset used in this paper is based on our previously published work on ADS-B attack detection using deep learning models [15]. In our prior work, the dataset generation process was only briefly described. To expand the transparency and reproducibility, this paper provides explanations of ADS-B attack scenarios and their implementation within the OpenScope environment. The dataset is publicly available here [54] and consists of:

- **malicious** messages generated using the enhanced version of OpenScope simulation platform by injecting controlled attack behaviors [55].
- **benign messages** collected from the OpenSky Network (real-world ADS-B data) [49].

A total of nine ADS-B attacks are considered, including message injection, spoofing, trajectory modification, message delay, and related message integrity and availability attacks. Figs. 5, 6 and 7 visually illustrate the attack generation process and dataset characteristics, which were not presented in detail in our earlier publication.

Fig. 5 visualizes the implementation of the nine selected ADS-B attack scenarios within the OpenScope simulator. The simulated environment presents a real-time visualization of aircraft movements, including flight trajectories, positions, and associated flight parameters. In the simulated environment, attacks are introduced by manipulating specific attributes of ADS-B messages, including aircraft position, altitude, heading, and transponder code.

Fig. 5 demonstrates how attack scenarios are injected into the simulated airspace environment, allowing controlled experimentation with different types of anomalous behavior. This visual representation highlights the interaction between benign and malicious aircraft behavior within a realistic air traffic control environment.

Fig. 6 provides a comprehensive visualization of attack injections in the OpenScope simulator, focusing on how specific fields of the ADS-B message are modified to induce malicious behavior. The simulated environment also provides precise control over aircraft attributes, allowing the creation of the selected attack types. The attacks are generated by modifying key parameters, including latitude, longitude, velocity, or the identity of the ADS-B message. The modification directly affects how the aircraft appears to other entities in the airspace. This controlled manipulation guarantees that the created malicious data closely mimics realistic attack patterns while maintaining consistency for experimental evaluation.

Fig. 7 illustrates the workflow of the attack-generation process used to generate the malicious dataset. The attack generation process begins with the collection of benign ADS-B messages, followed by selecting the corresponding attack type based on identified ADS-B vulnerabilities. For each attack generation, specific message attributes are modified in the OpenScope simulator's backend to replicate realistic adversarial behavior. The malicious are then validated and labeled according to

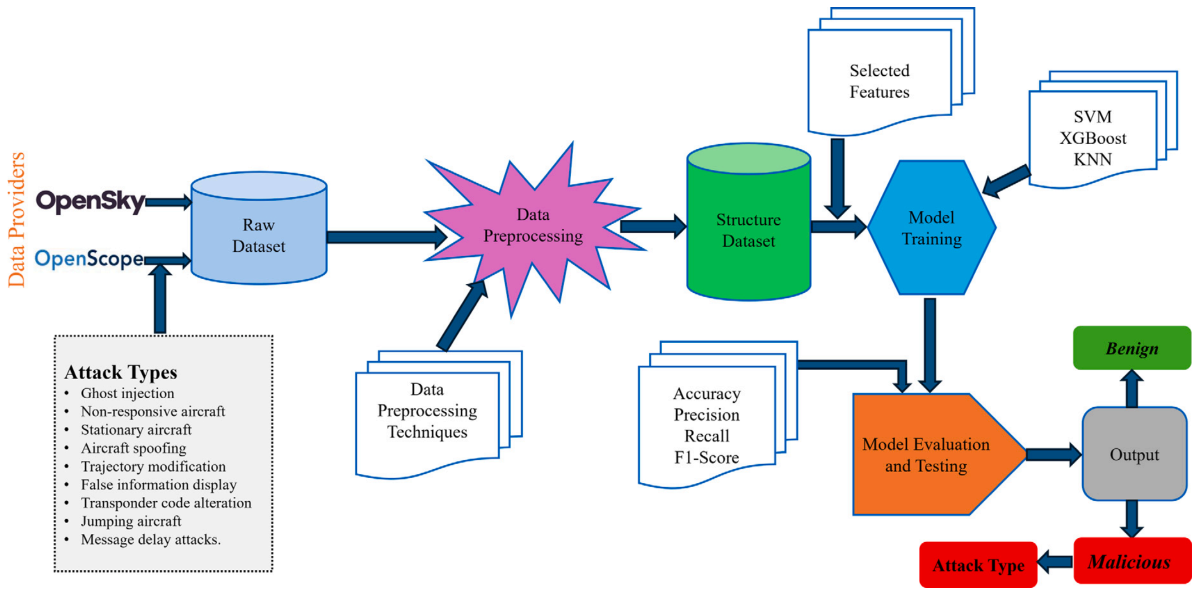

Fig. 3. Research methodology of the developed IDS (from data input to output).

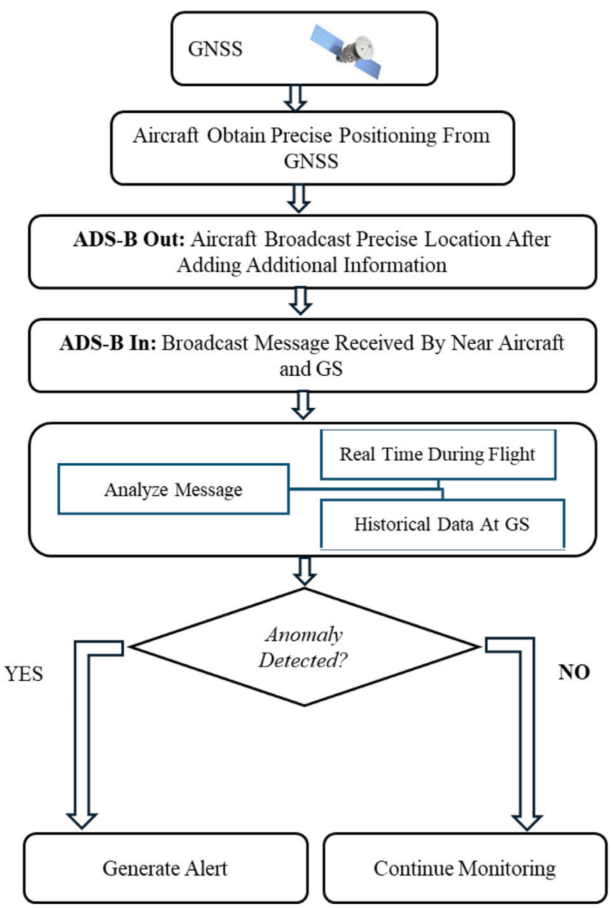

Fig. 4. Developed model flow chart.

Fig. 5. OpenScope simulation environment illustrating the implementation of ADS-B attack scenarios through manipulation of aircraft parameters.

Fig. 6. Detailed view of attack injection in OpenScope showing manipulation of ADS-B message attributes for generating malicious scenarios.

their respective attack categories. Finally, the dataset is combined and ready for use in ML model training and evaluation. The proposed structured methodology ensures consistency, reproducibility, and coverage of multiple attack scenarios within the dataset.

Dataset authenticity and limitations

Generating real-world ADS-B attack data in operational situations is highly constrained by safety, ethical, and regulatory considerations.

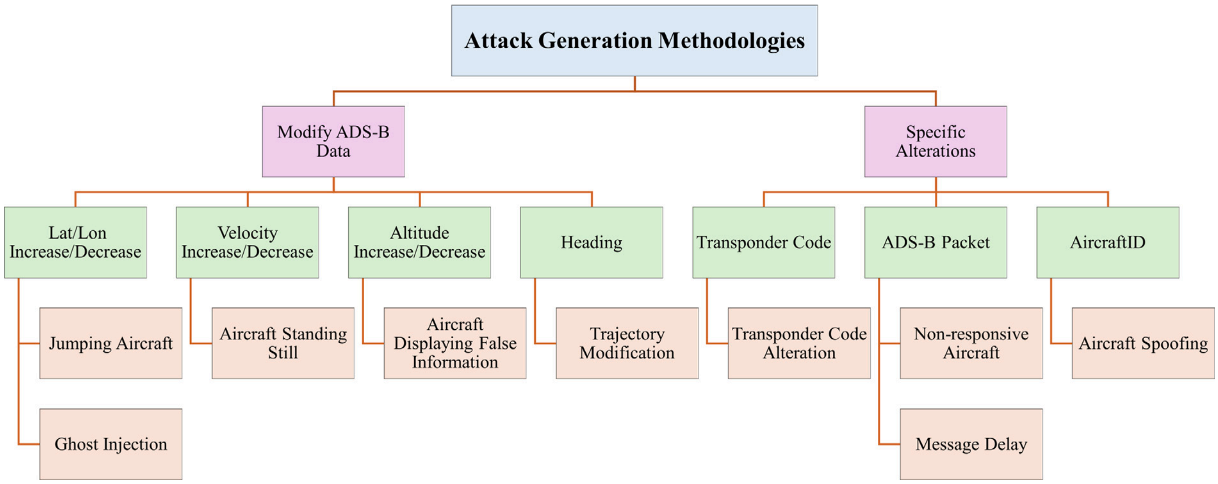

Fig. 7. Flowchart of the proposed attack generation methodology for constructing malicious ADS-B messages.

Conducting such attacks in operational situations could cause serious disruptions to air traffic systems. On the other hand, datasets available online contain extremely limited real-world ADS-B attack scenarios. To address this gap, this paper uses a hybrid dataset: benign messages are collected from the OpenSky Network to capture real air traffic behavior, while malicious are generated using the OpenScope simulation environment. The OpenScope simulator replicates real-world airspace dynamics, supporting controlled and physically plausible attack generation across multiple scenarios. Although simulation-based attack-generation methodologies might not fully capture the complexities of real adversarial behavior, they are still widely used in ADS-B security research and offer a practical, safe way to evaluate intrusion detection systems.

5.2. Data preprocessing

The initial dataset contained 20 attributes per ADS-B message, which were preprocessed to ensure suitability for machine learning model training. The feature engineering strategy employed in this study is grounded in aviation domain knowledge and the structural characteristics of ADS-B messages. The selected features primarily capture aircraft kinematics and spatial behavior, including position (latitude, longitude), velocity-related attributes (true airspeed, ground speed, speed), altitude, heading, and transponder information.

These features are specifically chosen to reflect physical flight dynamics and to enable the detection of anomalies introduced by malicious behavior. The effectiveness of this domain-driven feature design is validated through consistently high classification performance across multiple models, particularly KNN and XGBoost.

Further analysis using feature importance from the XGBoost model confirms that velocity-related features (e.g., *trueAirSpeed*, *groundSpeed*, and *speed*) are the most influential in distinguishing between normal and anomalous flight patterns. This demonstrates that the selected feature set captures the critical characteristics required for accurate intrusion detection without the need for extensive automated feature selection techniques.

5.3. Class imbalance mitigation strategies

To systematically address the inherent class imbalance in ADS-B telemetry datasets, targeted imbalance-aware learning strategies were developed and the baseline models were enhanced as follows:

- **KNN:** The neighbor-based classifier incorporates distance-weighted voting (`weights='distance'`). This ensures that closer minority-class instances exert greater influence during local neighborhood classification, mitigating the dominant voting bias of widespread majority classes.
- **SVM:** The boundary-based framework utilizes a linear support vector classifier (`LinearSVC`) configured with cost-sensitive learning (`class_weight='balanced'`). To capture complex non-linear attack distributions without incurring extreme computational latency, a Radial Basis Function (RBF) kernel approximation is implemented via the Nystroem transformation prior to linear classification. This configuration automatically penalizes misclassifications inversely proportional to class frequencies.
- **XGBoost:** The gradient-boosted tree architecture integrates sample-level weighting during the boosting iterations. Weights are computed dynamically using the inverse class frequency distribution, scaling the loss function gradients to ensure minority attack vectors contribute significantly to structural tree splits without destabilizing global model convergence.

5.4. Evaluation metrics

The performance of the proposed IDS was evaluated using standard classification metrics, considering both overall detection and multi-class attack identification. Accuracy (A), Precision (P), Recall (R), and F1-score were used to assess classification correctness, class-wise reliability, and the balance between precision and recall.

To provide a more robust evaluation under class imbalance, the Matthews Correlation Coefficient (MCC) was included as a balanced measure of overall classification quality. In addition, Receiver Operating Characteristic Area Under the Curve (ROC-AUC) and Precision-Recall Area Under the Curve (PR-AUC) were computed to evaluate the models' discriminative ability and performance on minority attack classes. These metrics offer deeper insight into model behavior beyond accuracy, particularly in imbalanced and safety-critical aviation scenarios.

6. Experiments and results

The experiments were conducted using Google Colab, a web-based platform that provides access to Google-provided graphical processing units (GPUs) and offers computational resources for efficient model training. The dataset was split into two parts: 80% for training and 20% for testing.

The complete pipeline and reproducibility resources are publicly available in the Zenodo repository [56]. It includes detailed preprocessing procedures, feature engineering steps, random seeds, software and library versions, model hyperparameters, training and testing datasets,

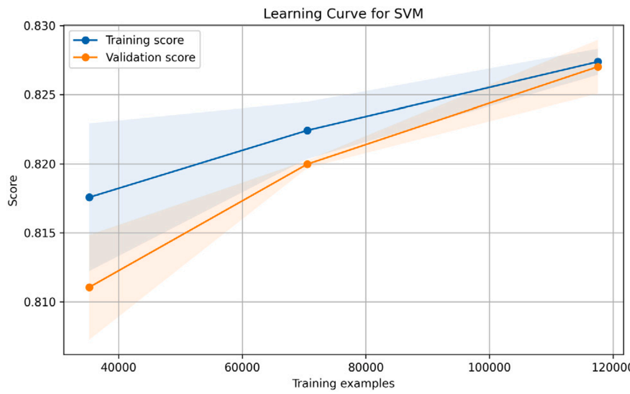

Fig. 8. Learning curve of the SVM model showing training and validation performance as a function of training size.

classification reports, runtime evaluation files, feature importance results, and the complete implementation code used for model training and evaluation. This repository enables full reproducibility of the experimental results presented in this study.

6.1. Support vector machine (SVM)

SVM is a flexible and robust model suitable for both binary and multi-class classification. It utilizes kernel methods to transform data into higher dimensions, allowing for effective separation even when the decision boundaries are not linear. The main objective of SVM in ADS-B message classification is to find the optimal hyperplane that best separates data points from one category from those of another. Support vectors are key data points nearest to the hyperplane and are essential for determining its location and orientation. These vectors are crucial for maximizing separation between classes, thereby ensuring robust classification.

For solving a binary classification problem using SVM, using a training dataset $\{(x_1, y_1), (x_2, y_2), \ldots, (x_n, y_n)\}$ where x_i denoted the feature vector and y_i denoted label of the class ($y_i \in \{-1, 1\}$), the algorithm attempts to find a suitable hyperplane defined in Eq. (1):

$$w \cdot x + b = 0 \tag{1}$$

Where:

- w is the weight vector.
- x is the input feature vector and
- b is the bias term.

The SVM algorithm aims to identify ideal values of w and b that increase the margin between the two classes. This is achievable by solving Eq. (2):

$$min_{w,b} \frac{1}{2} \|w\|^2 \tag{2}$$

Subject to the constraints (Eq. (3)):

$$y_i(w \cdot x_i + b) \geq 1 \quad \text{for all } i = 1, 2, \ldots, n \tag{3}$$

The SVM model achieves an overall accuracy of 0.83, with macro precision, recall, and F1-score of 0.64, 0.78, and 0.69, respectively. The weighted precision, recall, and F1-score are 0.87, 0.83, and 0.84, indicating reasonable performance on dominant classes but limited effectiveness on minority attack types. The gap between macro and weighted scores reflects class imbalance sensitivity.

The learning behavior (Fig. 8) shows gradual improvement with increasing data. The training score increases from approximately 0.817 at 30k samples to 0.827 at 120k samples, while the validation score improves from 0.811 to 0.827. The gap between the two curves narrows to near zero at higher sample sizes, indicating reduced variance. However, both curves plateau at a relatively lower performance level, suggesting underfitting and limited model capacity to capture complex class boundaries.

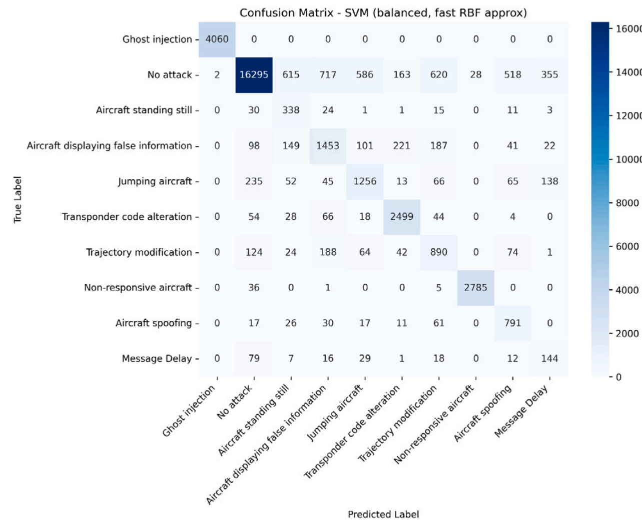

Fig. 9. Confusion matrix of the SVM model for multi-class ADS-B anomaly detection.

The confusion matrix (Fig. 9) reveals substantial misclassification, particularly for the *No attack* class. Although 16,295 instances are correctly classified, a significant number are confused with attack classes, including *False information* (717), *Trajectory modification* (620), *Aircraft standing still* (615), and *Jumping aircraft* (586). This indicates difficulty in separating benign and anomalous patterns.

Table 3 further highlights class-wise performance disparities. Perfect classification is achieved for *Ghost injection* (F1 = 1.00) and near-perfect results for *Non-responsive aircraft* (F1 = 0.99). Moderate performance is observed for *Transponder code alteration* (F1 = 0.87) and *No attack* (F1 = 0.88). In contrast, several classes exhibit weak detection capability, including *Aircraft standing still* (F1 = 0.41), *False information* (F1 = 0.58), and *Trajectory modification* (F1 = 0.57). The poorest performance is observed for *Message delay* (F1 = 0.30), indicating strong overlap with benign traffic.

Overall, the macro F1-score of 0.69 confirms limited balanced performance across classes. The results suggest that while SVM can capture clear attack patterns, it struggles with subtle and overlapping behaviors, leading to reduced generalization in complex multi-class settings.

6.2. Extreme gradient boosting (XGBoost)

XGBoost is a scalable algorithm intended for classification, ranking, and regression applications. Its foundation is an ensemble approach that builds a strong predictive model by combining the predictions of weak learners, usually decision trees. Each newly added tree in this process focuses on correcting errors introduced by previously added trees. The term “extreme” in XGBoost refers to the system improvements and algorithm refinements that make it highly effective and scalable for complex problems and large datasets. The objective function is responsible for combining the regularization terms and loss function represented in Eq. (4):

$$[\text{Obj} = \sum_{i=1}^{n} L(y_i, \hat{y}_i) + \sum_{k=1}^{T} \Omega(f_k)] \tag{4}$$

- **Loss Function** $L(y_i, \hat{y}_i)$: Calculate the error between the actual and predicted values.

Table 3
Classification performance of SVM model.

Class	Precision	Recall	F1-score
Ghost injection	1.00	1.00	1.00
No attack	0.96	0.81	0.88
Aircraft standing still	0.28	0.73	0.41
Aircraft displaying false information	0.56	0.61	0.58
Jumping aircraft	0.58	0.66	0.62
Transponder code alteration	0.83	0.91	0.87
Trajectory modification	0.53	0.62	0.57
Non-responsive aircraft	0.99	0.98	0.99
Aircraft spoofing	0.54	0.77	0.64
Message delay	0.22	0.48	0.30
Accuracy	0.83	0.83	0.83
Macro Avg	0.64	0.78	0.69
Weighted Avg	0.87	0.83	0.84

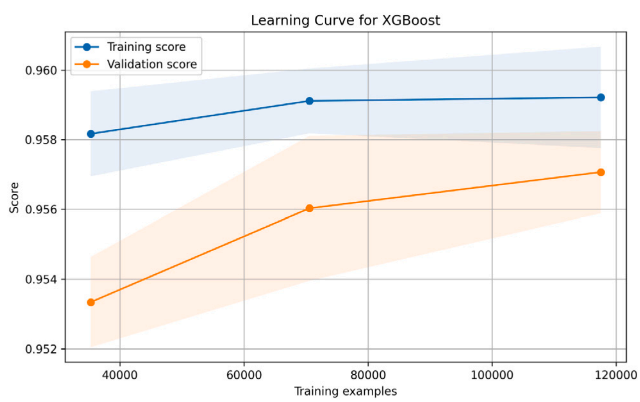

Fig. 10. Learning curve of the XGBoost model showing stable convergence and strong generalization across increasing training sizes.

- **Regularization Term** $\Omega(f_k)$: Boosts the algorithm to be generalizable and simple by penalizing the complexity of individual trees in the ensemble.

Eq. (5) represents the criterion for splitting nodes to maximize the gain:

$$G = \frac{1}{2}\left[\frac{\left(\sum_{i\in L} g_i\right)^2}{\sum_{i\in L} h_i + \lambda} + \frac{\left(\sum_{i\in R} g_i\right)^2}{\sum_{i\in R} h_i + \lambda} - \frac{\left(\sum_{i\in \text{parent}} g_i\right)^2}{\sum_{i\in \text{parent}} h_i + \lambda}\right] - \gamma \tag{5}$$

- **Maximizing Gain** G: The aim is to select splits that improve the model performance and the highest possible gain at each step.

Eq. (6) represents the final prediction for a data point by summing the outputs of all trees:

$$hat y(x) = \sum_{k=1}^{T} f_k(x) \tag{6}$$

The tuned XGBoost model achieves an accuracy of 0.98, with macro precision, recall, and F1-score of 0.92, 0.98, and 0.95, respectively. The weighted scores (0.9819 precision, 0.9791 recall, and 0.9799 F1-score) indicate consistent performance across imbalanced classes. The MCC of 0.9697 further confirms strong predictive reliability.

The learning behavior (Fig. 10) shows stable convergence. The training score increases marginally from 0.958 to 0.959 as the dataset grows from 30k to 120k samples, while the validation score improves from 0.953 to 0.957, indicating low variance. The difference between validation scores and final test accuracy is expected, as the learning curve is computed via cross-validation on partial training subsets.

Table 4
Classification performance of XGBoost model.

Class	Precision	Recall	F1-score
Ghost injection	1.00	1.00	1.00
No attack	1.00	0.97	0.98
Aircraft standing still	0.89	0.98	0.93
Aircraft displaying false information	0.93	0.99	0.96
Jumping aircraft	0.92	0.98	0.95
Transponder code alteration	1.00	1.00	1.00
Trajectory modification	0.94	1.00	0.97
Non-responsive aircraft	0.99	1.00	0.99
Aircraft spoofing	0.97	0.99	0.98
Message Delay	0.58	0.89	0.70
Accuracy	0.98	0.98	0.98
Macro Avg	0.92	0.98	0.95
Weighted Avg	0.98	0.98	0.98

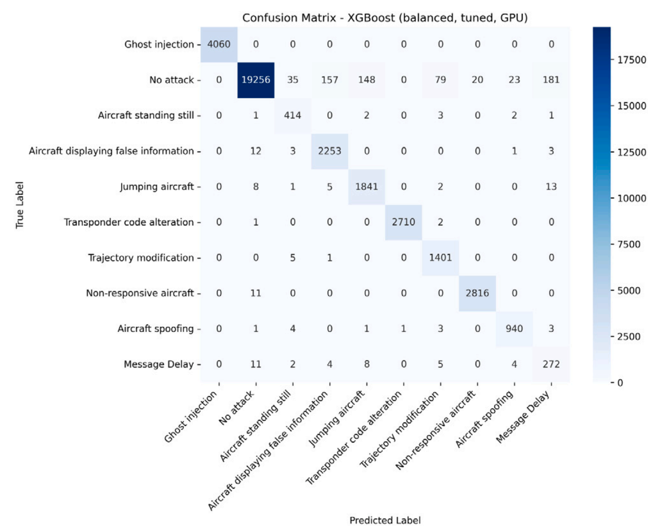

Fig. 11. Confusion matrix of the XGBoost model demonstrating high classification accuracy across all ADS-B attack classes.

The confusion matrix (Fig. 11) demonstrates near-perfect class separability. *Ghost injection* is classified without error (4060/4060). The *No attack* class achieves 19,256 correct predictions with limited confusion, mainly toward *False information* (157), *Jumping aircraft* (148), and *Message delay* (181). Minority classes exhibit near-complete recall, including *Transponder code alteration* (2710/2713), *Trajectory modification* (1401/1406), and *Non-responsive aircraft* (2816/2827).

Table 4 further quantifies class-wise performance. Perfect scores are observed for *Ghost injection* and *Transponder code alteration* (precision = recall = F1 = 1.00). High performance is also achieved for *Non-responsive aircraft* (F1 = 0.99), *Aircraft spoofing* (F1 = 0.98), and *Trajectory modification* (F1 = 0.97). Moderate degradation appears in relatively harder classes such as *Aircraft standing still* (F1 = 0.93) and *False information* (F1 = 0.96), primarily due to minor overlap with benign patterns. The lowest performance is observed for *Message delay* (F1 = 0.70), reflecting its ambiguity and similarity to normal traffic.

Overall, the macro F1-score of 0.95 indicates strong balanced performance across all classes, while the weighted F1-score of 0.98 confirms robustness under class imbalance.

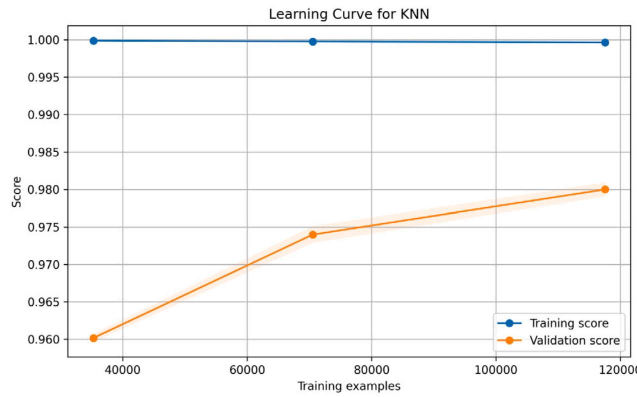

Fig. 12. Learning curve of the KNN model showing near-perfect training performance and steadily improving validation accuracy as the training set increases.

6.3. k-Nearest neighbors (KNN)

KNN is a simple, non-parametric algorithm used for addressing classification and regression tasks. It functions based on the notion that similar instances tend to yield similar results. Rather than learning a model during training, KNN retains the training dataset and predicts outcomes by assessing the similarity between a query point and the training data. In the KNN algorithm, the initial step is to choose a value for k, which decides the number of training points that will affect the prediction for a given query point. For a specified query point, KNN determines the distance from this point to every point in the training dataset. The Euclidean distance is most frequently utilized, although other metrics may also be used. Using the computed distances, the algorithm finds the k nearest data points (neighbors) to the query point.

Eq. (7) represents the mathematical form of Euclidean distance between two points $A = (a_1, a_2, \dots, a_n)$ and $B = (b_1, b_2, \dots, b_n)$ in an n-dimensional space. In comparison, Equation 11 represents the mathematical form of classification after calculating the distance between two points; based on the majority class among the k nearest neighbors, the class label C is predicted.

$$d(A, B) = \sqrt{\sum_{i=1}^{n} (a_i - b_i)^2} \tag{7}$$

Where:

- $A = (a_1, a_2, \dots, a_n)$ represents the feature vector of the query point.
- $B = (b_1, b_2, \dots, b_n)$ represents the feature vector of a training point.
- $d(A, B)$ represents the calculated Euclidean distance between A and B (Eq. (8)).

$$\hat{C} = \text{mode}(C_1, C_2, \dots, C_k) \tag{8}$$

Where:

- $C_1, C_2, \dots, C_k$ are the classes of the k nearest neighbors.
- $\hat{C}$ is the predicted class label for the query point.

The KNN model achieves an overall accuracy of 0.98, with macro precision, recall, and F1-score of 0.96, 0.94, and 0.95, respectively. The weighted precision, recall, and F1-score are all 0.98, indicating consistently strong performance across both majority and minority classes. Compared to SVM, the reduced gap between macro and weighted metrics suggests improved handling of class imbalance.

The learning behavior (Fig. 12) shows near-perfect training performance across all dataset sizes, with the training score remaining at approximately 0.99. The validation score increases steadily from 0.96 at 30k samples to 0.98 at 120k samples. While the persistent gap between training and validation curves indicates mild overfitting, the continuous improvement of the validation score suggests that the model benefits from additional data and maintains strong generalization.

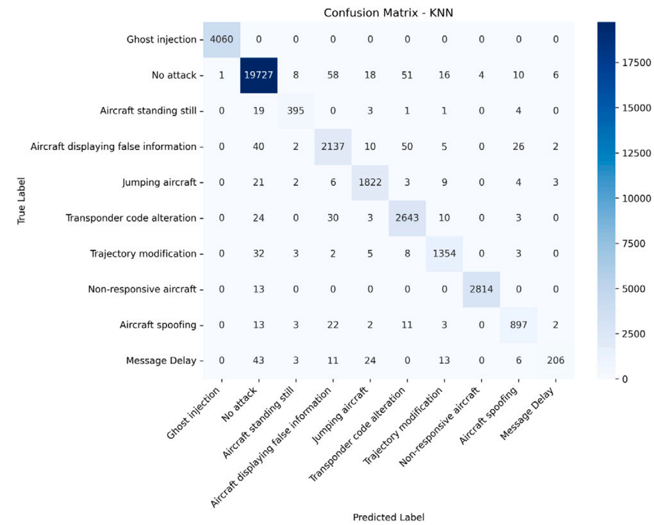

Fig. 13. Confusion matrix of the KNN model for multi-class ADS-B anomaly detection, demonstrating strong class-wise discrimination with limited confusion among minority attack classes.

The confusion matrix (Fig. 13) demonstrates high classification accuracy across most classes. The *Ghost injection* class is perfectly classified (4060/4060), and *Non-responsive aircraft* also achieves near-perfect detection (2814 correct predictions). The *No attack* class shows strong performance with 19,727 correct predictions, with only minor confusion distributed across attack categories such as *False information* (58) and *Transponder alteration* (51).

Table 5 further confirms robust class-wise performance. High F1-scores are observed for most classes, including *Jumping aircraft* (0.97), *Transponder code alteration* (0.96), and *Trajectory modification* (0.96). The *Aircraft standing still* and *Aircraft displaying false information* classes also show strong balance between precision and recall (F1 = 0.94).

However, performance degradation is observed for the *Message delay* class, which achieves a lower recall of 0.67 and F1-score of 0.78, indicating residual confusion with benign and temporally similar patterns. This suggests that KNN, while effective in capturing local structure, may struggle with classes exhibiting temporal overlap or subtle feature differences.

Overall, the macro F1-score of 0.95 indicates strong balanced performance across classes. The results demonstrate that KNN provides competitive accuracy and class-wise consistency, although slight overfitting and sensitivity to overlapping patterns remain observable limitations.

7. Discussion

7.1. Performance comparison of the models

The comparative results of the evaluated models are summarized in Table 6 and illustrated in Fig. 14. Overall, KNN and XGBoost significantly outperform SVM across all evaluation metrics, achieving identical accuracy (0.98), substantially higher than SVM (0.83), thereby demonstrating superior predictive capability.

From a class-balanced perspective, both KNN and XGBoost achieve strong macro F1-scores (0.95), whereas SVM shows significantly lower performance (0.69), indicating poor handling of minority classes. While XGBoost attains the highest macro recall (0.98), KNN demonstrates slightly higher macro precision (0.96), suggesting more conservative

Table 5
Classification performance of KNN model.

Class	Precision	Recall	F1-score
Ghost injection	1.00	1.00	1.00
No attack	0.99	0.99	0.99
Aircraft standing still	0.95	0.93	0.94
Aircraft displaying false information	0.94	0.94	0.94
Jumping aircraft	0.97	0.97	0.97
Transponder code alteration	0.96	0.97	0.96
Trajectory modification	0.96	0.96	0.96
Non-responsive aircraft	1.00	1.00	1.00
Aircraft spoofing	0.94	0.94	0.94
Message Delay	0.94	0.67	0.78
Accuracy	0.98	0.98	0.98
Macro Avg	0.96	0.94	0.95
Weighted Avg	0.98	0.98	0.98

Table 6
Quantitative comparison of KNN, SVM, and XGBoost models across performance metrics, robustness measures, and computational efficiency.

Metric	KNN	SVM	XGBoost
Accuracy	0.98	0.83	0.98
Precision (Macro)	0.96	0.65	0.92
Recall (Macro)	0.94	0.78	0.98
F1-score (Macro)	0.95	0.69	0.95
Precision (Weighted)	0.98	0.87	0.98
Recall (Weighted)	0.98	0.83	0.98
F1-score (Weighted)	0.98	0.84	0.98
MCC	0.97	0.77	0.97
Training Time (s)	0.40	657.97	22.22
Testing Time (s)	1.17	0.18	0.48
ROC-AUC (Macro)	0.98	0.98	1.00
PR-AUC (Macro)	0.95	0.75	0.98
ROC-AUC (Weighted)	0.99	0.97	1.00
PR-AUC (Weighted)	0.98	0.91	1.00

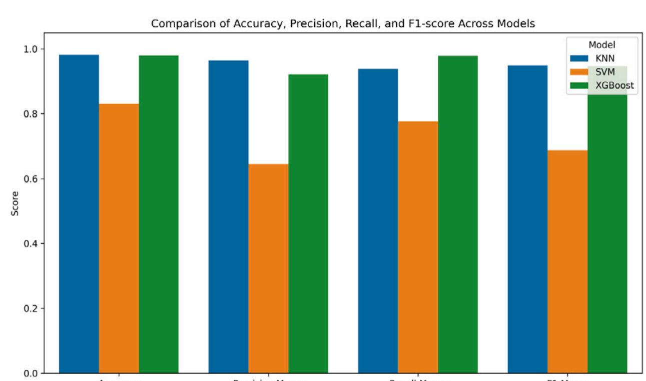

Fig. 14. Comparison of accuracy, macro precision, macro recall, and macro F1-score across KNN, SVM, and XGBoost models, highlighting the competitive performance of KNN and XGBoost relative to SVM.

and reliable predictions with fewer false positives—an important property for real-time anomaly detection systems.

In terms of robustness, KNN and XGBoost achieve comparable MCC values (0.97), confirming stable and consistent predictions, whereas SVM remains considerably lower (0.77). Although XGBoost achieves near-perfect ROC-AUC and PR-AUC scores, the marginal improvement over KNN does not translate into a significant practical advantage.

Table 7
Performance comparison on selected minority attack classes using ROC-AUC and PR-AUC metrics.

Model	Class name	ROC-AUC	PR-AUC
KNN	Message Delay	0.88	0.73
KNN	Aircraft standing still	0.98	0.94
KNN	Aircraft spoofing	0.99	0.95
SVM	Message Delay	0.94	0.36
SVM	Aircraft standing still	0.98	0.53
SVM	Aircraft spoofing	0.98	0.67
XGBoost	Message Delay	0.99	0.88
XGBoost	Aircraft standing still	1.00	0.99
XGBoost	Aircraft spoofing	1.00	0.99

From a computational perspective, KNN provides a distinct advantage for real-time deployment, with an extremely low training time (0.4 s) and a simple, non-parametric structure. While KNN exhibits slightly higher inference time than SVM, its deployment simplicity, absence of complex model training, and adaptability to streaming data make it highly suitable for real-time systems. In contrast, SVM incurs prohibitively high training cost (657.97 s), and although XGBoost offers balanced performance (22.22 s training), it introduces additional complexity due to model tuning and GPU dependency.

7.2. Performance on minority classes

To further evaluate model robustness, performance on minority classes was analyzed using per-class precision, recall, F1-score, and additional threshold-independent metrics such as ROC-AUC and PR-AUC. While overall accuracy remains high, class-wise analysis reveals that minority attack categories exhibit comparatively lower recall, indicating challenges in detecting less frequent anomalies.

Table 7 presents the ROC-AUC and PR-AUC scores for selected minority attack classes across the evaluated models. The results show that XGBoost consistently achieves near-perfect performance, with ROC-AUC values reaching 1.00 and PR-AUC values up to 0.99, indicating excellent separability and precision–recall balance even under class imbalance.

The KNN model demonstrates strong and stable performance across minority classes, achieving high ROC-AUC scores (up to 0.99) and competitive PR-AUC values. In contrast, SVM exhibits noticeable degradation in PR-AUC, particularly for the Message Delay class (0.36), highlighting its sensitivity to class imbalance and difficulty in distinguishing subtle anomalies.

These findings reinforce that KNN maintains robustness under imbalanced conditions while offering computational simplicity, making it suitable for real-time anomaly detection scenarios where rare events are critical. Furthermore, the consistently lower performance for Message Delay across all models confirms its inherent complexity and overlap with benign traffic patterns.

7.3. Ablation analysis and performance trade-offs

To improve transparency regarding the effect of imbalance-aware learning, an ablation study was performed by comparing class-unbalanced (un-optimized) and class-balanced (optimized) model configurations, as presented in Table 8. The baseline configurations rely on default algorithmic setups, whereas the optimized implementations incorporate targeted hyperparameter tuning, distance-weighted voting (`weights='distance'`) for KNN, balanced class weighting for SVM (`class_weight='balanced'`), and sample-level weighting for XGBoost.

The empirical results show that class-balanced learning significantly improved minority-class detection and macro-level robustness across

Table 8
Ablation analysis comparing un-optimized (class-unbalanced) and optimized (class-balanced) implementations of KNN, SVM, and XGBoost.

Metric	Un-Optimized (Class-unbalanced)			Optimized (Class-balanced)		
	KNN	SVM	XGBoost	KNN	SVM	XGBoost
Accuracy	0.99	0.85	0.94	0.98	0.83	0.98
Precision (Macro)	0.98	0.82	0.96	0.96	0.65	0.92
Recall (Macro)	0.96	0.55	0.79	0.94	0.78	0.98
F1-score (Macro)	0.96	0.60	0.85	0.95	0.69	0.95
Precision (Weighted)	0.99	0.84	0.94	0.98	0.87	0.98
Recall (Weighted)	0.99	0.85	0.94	0.98	0.83	0.98
F1-score (Weighted)	0.99	0.83	0.93	0.98	0.84	0.98
MCC	0.98	0.78	0.91	0.97	0.77	0.97
Training Time (s)	0.41	928.50	4.81	0.40	657.97	22.22
Testing Time (s)	0.52	270.82	0.05	1.17	0.18	0.48
ROC-AUC (Macro)	0.99	0.92	0.99	0.98	0.98	1.00
PR-AUC (Macro)	0.98	0.60	0.91	0.95	0.75	0.98
ROC-AUC (Weighted)	1.00	0.93	0.99	0.99	0.97	1.00
PR-AUC (Weighted)	0.99	0.82	0.97	0.98	0.91	1.00

the models, particularly for SVM and XGBoost. For the SVM framework, the macro recall increased from 0.55 to 0.78, the macro F1-score rose from 0.60 to 0.69, and the macro PR-AUC advanced from 0.60 to 0.75. Notably, the Nystroem transformation not only dropped testing latency from 270.82 s to 0.18 s, but also reduced training overhead from 928.50 s to 657.97 s by mapping the kernel space into a computationally efficient linear alternative.

XGBoost achieved the most substantial optimization gains; its overall classification accuracy rose from 94.00% to 98.00%, while its macro recall experienced a stark increase from 0.79 to 0.98, and its macro F1-score advanced from 0.85 to 0.95. By tracking comprehensive metrics like the Matthews Correlation Coefficient (MCC) and macro-averaged AUCs across both columns, the ablation framework clearly reveals that while un-optimized models display deceptively high global accuracies due to majority-class dominance, the proposed adjustments successfully restore classification balance without compromising real-time operational feasibility.

Conversely, certain metrics experienced marginal declines in the optimized configurations, illustrating the classical precision–recall trade-off inherent to cost-sensitive learning. For instance, the SVM configuration saw its macro precision drop from 0.82 to 0.65, while its global accuracy adjusted from 0.85 to 0.83. This shift occurs because the balance-weighting mechanism heavily penalizes minority-class omissions, shifting the decision boundaries to favor high recall on critical attack vectors. Consequently, while the model successfully raised its macro recall by 23 percentage points (from 0.55 to 0.78), it naturally incurred a higher false-alarm rate among overlapping majority-class instances. Similarly, the slight performance adjustments in the distance-weighted KNN underscore the sensitivity of local instance-based neighborhood boundaries to highly skewed, overlapping feature spaces. In a security-critical aviation environment, prioritizing a high detection rate (Recall) over absolute exactness (Precision) represents a necessary and justified design choice to prevent catastrophic miss-detections.

7.4. Analysis of weakly detected attack classes

Although the overall detection performance improves significantly after model tuning, a small number of attack classes remain comparatively challenging. In particular, the *Message Delay* class continues to exhibit lower recall and precision across models, even though substantial improvements are observed compared to earlier experiments. This difficulty arises from the fact that delayed messages preserve physically consistent spatial and kinematic properties (e.g., speed, heading, and position), while deviating primarily in temporal consistency. Since the current models operate on snapshot-based features, such temporal inconsistencies are not explicitly captured, leading to occasional misclassification as benign traffic.

In contrast to earlier results, previously challenging classes such as *Aircraft standing still* and *Aircraft spoofing* now achieve high precision and recall, particularly in KNN and XGBoost models. This indicates that improved feature representation and model tuning have significantly enhanced separability for spatially distinguishable anomalies.

The remaining misclassifications are sparse and do not indicate systematic bias but rather reflect inherent overlap between certain anomalous and normal flight behaviors. These findings suggest that while the current approach is highly effective, further improvements could be achieved by incorporating temporal or sequence-aware modeling techniques to better capture time-dependent anomalies such as message delays.

7.5. Comparison with state-of-the-art

Table 9 presents a comparative analysis of the proposed approach against recent ADS-B intrusion detection studies. Existing works predominantly rely on hybrid datasets combining real ADS-B data with synthetically generated attack vectors and typically focus on a limited set of attacks, such as message injection, spoofing, or trajectory manipulation.

In terms of detection performance, prior studies report accuracies ranging from 91% to 97.99%, including LSTM-based approaches (between 91% and 93.04%), RNN-GRU (94.61%), and FEPAN (97.99%). The IMO-ELM-GRNN framework achieves approximately 98% accuracy, representing one of the highest reported performances under similar experimental conditions. However, most of these approaches do not provide computational performance metrics, limiting their assessment for real-time deployment.

In contrast, the proposed work achieves competitive accuracy (KNN: 98.16%, XGBoost: 98%) while significantly extending the attack coverage to nine diverse and operationally relevant scenarios. More importantly, it explicitly evaluates computational efficiency, demonstrating substantially lower training times (0.40 s for KNN and 22.22 s for XGBoost) and fast inference (1.17 s and 0.48 s, respectively), compared to the limited available benchmarks (e.g., RNN-GRU with 53.2 s training time).

Overall, while existing studies emphasize detection accuracy under constrained attack settings, the proposed approach provides a more comprehensive and practically deployable solution by jointly addressing multi-class detection, dataset realism, and real-time feasibility.

7.6. Model interpretability and feature importance

To better understand the contribution of input variables to the classification process, feature importance analysis was conducted using the XGBoost model, as shown in Fig. 15. The results indicate

Table 9
Performance comparison of ADS-B intrusion detection approaches across datasets, attack types, and computational metrics.

Ref	Dataset	Attacks	Accuracy	Training time	Testing time
[21]	Simulated Dataset	Spoofing attack, Message injection attack, and Virtual trajectory attack	RCF: 96		
[25]	Real ADS-B data collection + Synthetically generated attack	Replay attack, Ghost aircraft injection attack, and Multiple ghost aircraft injection attack	SVM: 97%	–	–
[26]	OpenSky	Message injection attack	LSTM: 93.04%	–	–
[27]	Real ADS-B data collection + Synthetically generated attack	Injection attacks	RNN-GRU: 94.61%	53.2 s	0.7 s
[31]	Simulated Dataset	Spoofing attack	LSTM: 91%	–	–
[35]	Real data + Artificially generated attacks	Four scenario-based attacks	FEPAN: 97.99%	–	–
[36]	Real data + Artificially generated attacks	Message deletion, Message injection, Message modification	GAN-LSTM: 92.31%	–	–
[57]	Real ADS-B flight data + Artificially injected attacks	Message deletion, Message injection, Message modification	IMO-ELM-GRNN: 98%	–	–
Our Work	OpenSky and OpenScope	Ghost injection attack, Nonresponsive aircraft attack, Stationary aircraft attack, Aircraft spoofing attack, Trajectory modification attack, False information display attack, Transponder code alteration attack, Jumping aircraft attack, and Message delay attack	KNN: 98.16%, XGBoost: 98%	0.40 s, 22.22 s	1.17 s, 0.48 s

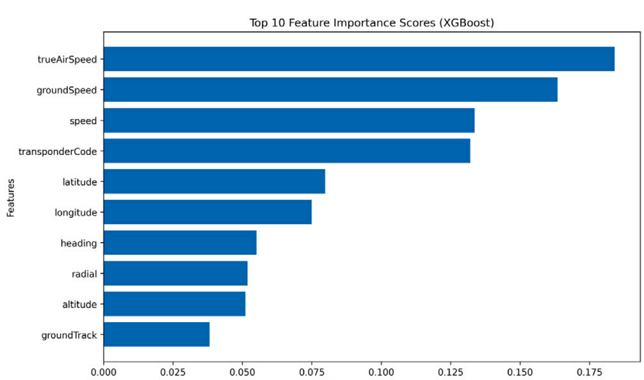

Fig. 15. Top 10 feature importance scores derived from the XGBoost model.

that *trueAirSpeed* and *groundSpeed* are the most influential features, followed by *speed* and *transponderCode*. These features are directly associated with aircraft motion dynamics, highlighting their critical role in distinguishing normal and anomalous flight behaviors.

Geospatial features such as *latitude* and *longitude* exhibit moderate importance, suggesting that positional context contributes to anomaly detection but is less dominant than velocity-related parameters. Other features, including *heading*, *radial*, and *altitude*, provide complementary information that enhances classification performance.

Although XGBoost is used for interpretability due to its inherent feature attribution capability, the insights gained are also applicable to KNN-based deployment, as both models rely on the same underlying feature space. This further supports the selection of KNN for real-time deployment while maintaining interpretability of the feature contributions.

7.7. Real-time deployment feasibility

To assess the practical applicability of the proposed IDS, the real-time performance of the KNN model was evaluated by measuring the average classification time per ADS-B message across varying batch sizes. KNN is selected for deployment due to its strong classification performance and consistent behavior under imbalanced conditions.

Fig. 16 presents the average processing time per message (in milliseconds) as a function of the number of input messages. The results show that after an initial increase, the classification latency stabilizes at approximately 5.4–5.6 ms per message, indicating consistent computational performance independent of batch size. This stability suggests that the model operates in a steady processing regime without significant degradation under increased workload.

An average latency of approximately 5.5 ms per message corresponds to a throughput of roughly 180 messages per second on a single CPU core. In operational ADS-B environments, each aircraft typically transmits around two messages per second. Therefore, a single-core implementation can support approximately 90 aircraft in real time.

While this single-core throughput is lower than ideal for dense airspace scenarios, modern deployment environments can leverage parallel processing. By distributing the workload across multiple CPU cores, the system can achieve near-linear scalability. For example, using eight cores increases throughput to approximately 1440 messages per second, enabling support for over 700 aircraft simultaneously, which exceeds typical high-density air traffic requirements.

It is important to note that real-time constraints in ADS-B systems are not strictly limited to sub-millisecond latency but rather depend on

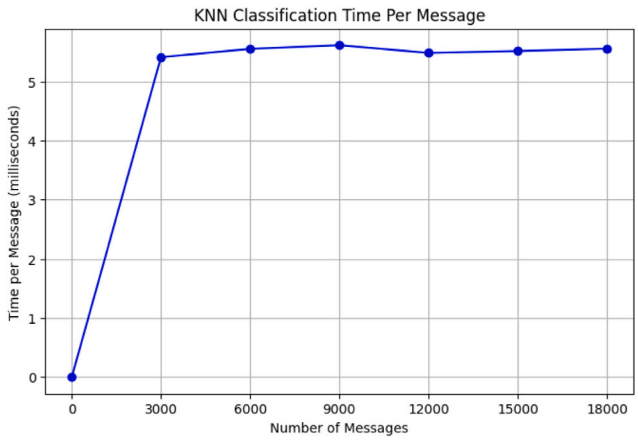

Fig. 16. Per-message classification latency of the KNN model across different batch sizes.

maintaining processing rates that match or exceed incoming message streams. In this context, the proposed KNN-based IDS satisfies real-time requirements when deployed in a parallel computing environment.

Overall, these results demonstrate that the KNN model provides a practical balance between detection performance and deployment feasibility, making it a suitable candidate for real-time intrusion detection in ADS-B systems.

8. Conclusion

ADS-B is a next-generation technology used for air traffic control and surveillance. By leveraging GPS, ADS-B provides precise positioning data and enhances aircraft tracking by broadcasting detailed information about each aircraft's location. Despite its many advantages, ADS-B remains susceptible to various security threats due to its open and unencrypted message transmission. Beyond achieving high detection accuracy, this study provides empirical insight into which ADS-B attack classes remain intrinsically difficult to detect using lightweight, per-message classifiers. The findings indicate that temporally subtle attacks, such as *message delay*, require sequence-aware modeling, while kinematically plausible manipulations pose challenges for traditional classifiers. These insights can directly inform the design of future EC-enabled surveillance architectures and guide the integration of IDS components within UTM–ATM environments.

This paper presents a comprehensive threat model for identifying and categorizing ADS-B attacks, with the objective of enhancing communication security. A novel dataset comprising nine distinct attack types is introduced, and three machine learning models – KNN, SVM, and XGBoost – are developed and evaluated for multi-class intrusion detection. These models are used to classify both benign and malicious messages, as well as to identify specific attack categories.

Experimental results demonstrate that both KNN and XGBoost achieve high classification performance, with an overall accuracy of 98%, significantly outperforming SVM (83%). KNN achieves strong macro-level performance (F1-score: 0.95) and high robustness (MCC: 0.97), while also offering efficient training and stable real-time inference characteristics. XGBoost provides comparable accuracy with superior recall and near-perfect ROC-AUC and PR-AUC scores, indicating strong discriminative capability. However, considering deployment constraints and real-time processing requirements, KNN offers a more practical balance between performance and computational efficiency.

Table 10
List of abbreviations.

Abbreviation	Description
ADS-B	Automatic Dependent Surveillance–Broadcast
ATC	Air Traffic Control
ATM	Air Traffic Management
CIA+A	Confidentiality, Integrity, Availability, Authentication
CNS	Communication, Navigation, and Surveillance
CPDLC	Controller Pilot Data Link Communication
DL	Deep Learning
DoS	Denial of Service
DDoS	Distributed Denial of Service
FAA	Federal Aviation Administration
FNR	False Negative Rate
FPR	False Positive Rate
GNSS	Global Navigation Satellite System
GPS	Global Positioning System
GS	Ground Station
GRU	Gated Recurrent Unit
IDS	Intrusion Detection System
ICAO	International Civil Aviation Organization
KNN	K-Nearest Neighbors
LSTM	Long Short-Term Memory
ML	Machine Learning
PR	Precision
RC	Recall
RF	Random Forest
ROC	Receiver Operating Characteristic
RNN	Recurrent Neural Network
SDR	Software-Defined Radio
SMOTE	Synthetic Minority Over-sampling Technique
SVM	Support Vector Machine
TCAS	Traffic Collision Avoidance System
UAT	Universal Access Transceiver
XAI	Explainable Artificial Intelligence
XGBoost	Extreme Gradient Boosting

8.1. Limitations and future work

In this paper, we did not examine the system's scalability in high-traffic scenarios; instead, we concentrate on assessing IDS performance using simple computational time. To combat evolving threats and assist IDS in dynamically adapting to novel attack patterns, our future work will focus on implementing online learning strategies such as reinforcement learning and incremental learning. To evaluate the IDS's performance in practical settings and ensure scalability, we will conduct extensive tests in high-traffic airspace scenarios. Table 10 illustrates the list of abbreviations used in the paper.

CRediT authorship contribution statement

Waqas Ahmed: Writing – original draft, Methodology, Investigation, Conceptualization. **Ammar Masood:** Validation, Supervision, Resources, Methodology. **Jawad Manzoor:** Writing – review & editing, Validation, Supervision.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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The authors utilized ChatGPT and Grammarly solely for language refinement, grammar correction, and improvement of manuscript clarity. All scientific contributions, including data collection, analysis, methodology, and interpretation of results, are entirely the original work of the authors. All figures and experimental results presented in this paper are also original and were generated without the use of AI-based image generation tools.

Data availability

All datasets, implemented models, and experimental results used in this study are publicly available at [56]. The generated raw ADS-B dataset is separately provided at [54]. These resources include the preprocessing pipeline, training/testing splits, and evaluation outputs, enabling full reproducibility of the reported results and facilitating reuse in future research.

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